



DGLRM

From Bed Rest to Mars:

Development of a Literature-Derived Machine Learning Framework for Predicting Lower-Limb Muscle Atrophy in Spaceflight Analogues

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64. Jahrestagung der DGLRM

Agenda

Part 1: background and data

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Motivation

Why forecast muscle loss, and why bed rest

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Data

Range of unloading, search and dataset

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Muscles

Which muscles lose most

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Models

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LLM forecast

What it sees, results, checks

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Findings, limits, next steps

The marker in the bottom-right corner of each slide shows where we are.



Motivation: can we predict muscle atrophy?

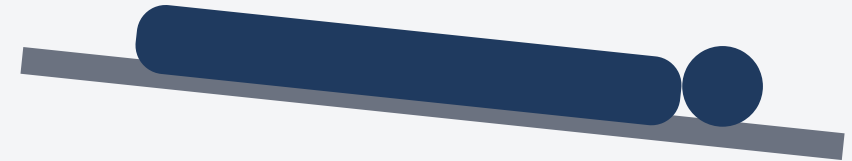
What are we doing?

- Lower-limb muscle atrophy is a major challenge in long-duration spaceflight
- Research question: can the extent of muscle atrophy be predicted?
- Aim: a literature-derived dataset and a machine-learning framework for this prediction

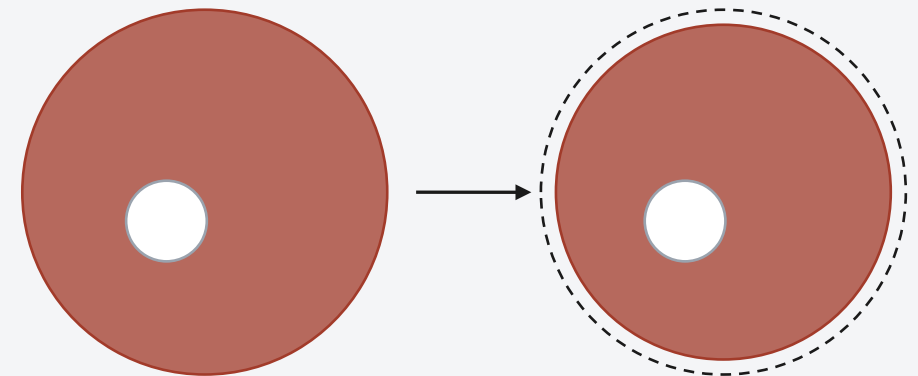
Why is this important?

- Prevention: identify risk early and leave time to intervene
- Countermeasures: match exercise and nutrition to the predicted risk
- Vulnerable muscles: show which muscles need closer monitoring
- Function: less muscle loss helps preserve strength and power
- Mission planning: estimate muscle loss on long missions with limited exercise equipment

Ground analogue: 6° head-down bed rest



Weeks to months in bed; legs carry no body weight



Before bed rest

Day 60: -16%

Calf cross-section to scale; dashed line = size before. Model estimate for control groups (plantar flexors, -15.8% at day 60).

Why from bed rest to Mars?

Bed rest: the ground analogue

- Direct spaceflight data on muscle are scarce
- Head-down bed rest is the established ground analogue: it unloads the legs under controlled conditions
- Not equivalent to spaceflight, but a sound basis to build and test prediction models

Mars: the long-term application

- A Mars mission lasts about 2.5 years, with about 6 months of transit each way
- No early return in an emergency: the crew depends on its own fitness
- After months in weightlessness, the crew must leave the spacecraft and work on the surface

Moon mission vs Mars mission



What range did we study?

Abstract
15 bed-rest studies

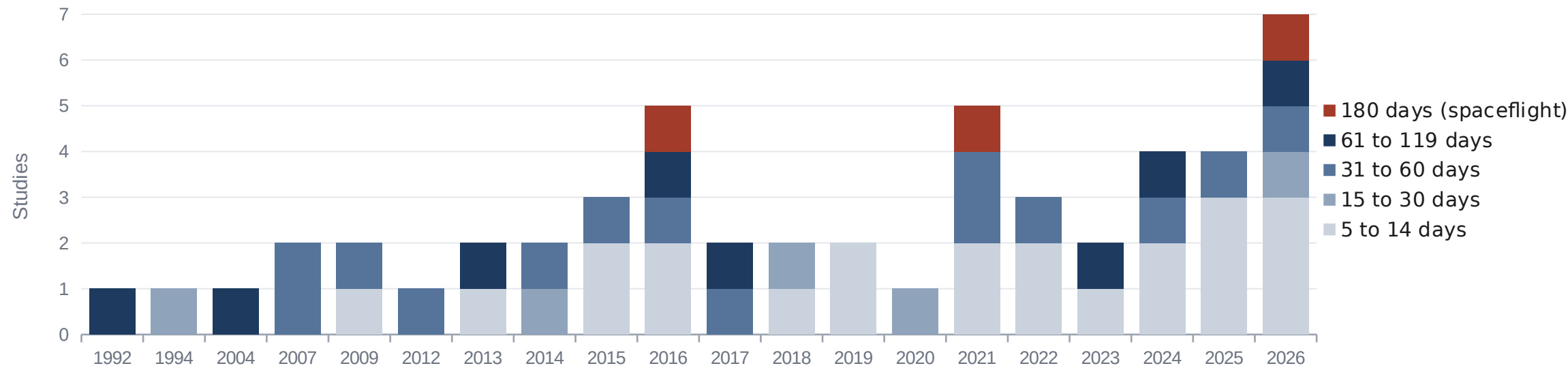


Final dataset
52 studies

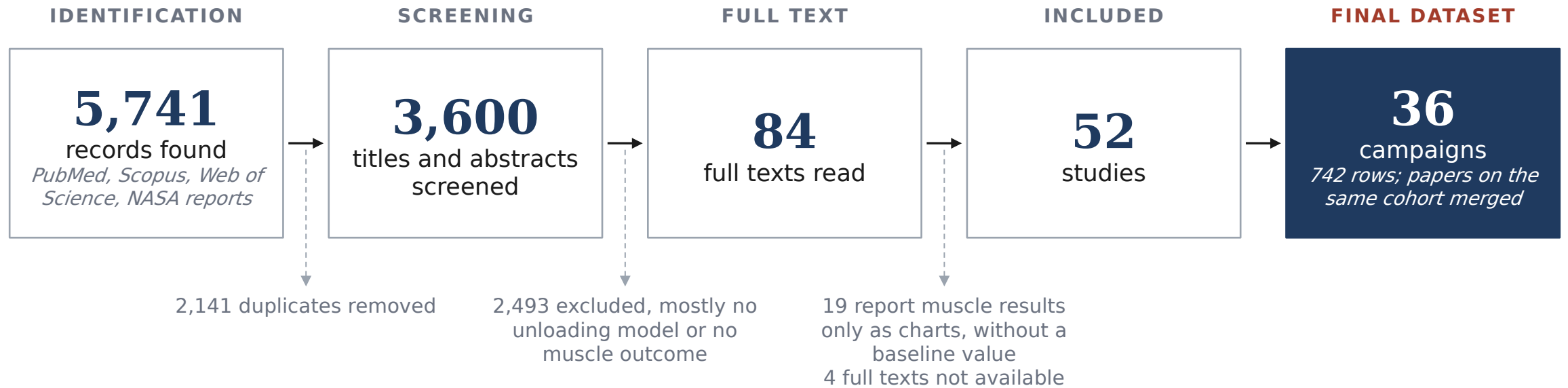


Days of unloading 0 30 60 90 119 150 180

Studies published per year, by length of unloading (years without studies omitted)



How did we build the dataset?



Flow of records in the style of PRISMA 2020, the standard for reporting systematic reviews.

What does the dataset look like?

Final dataset at a glance

Feature	n
Observations (rows)	742
Studies	52
Independent campaigns	36
Muscles or muscle groups	51
Rows: control / countermeasure	470 / 272
Rows: during bed rest / recovery	478 / 264

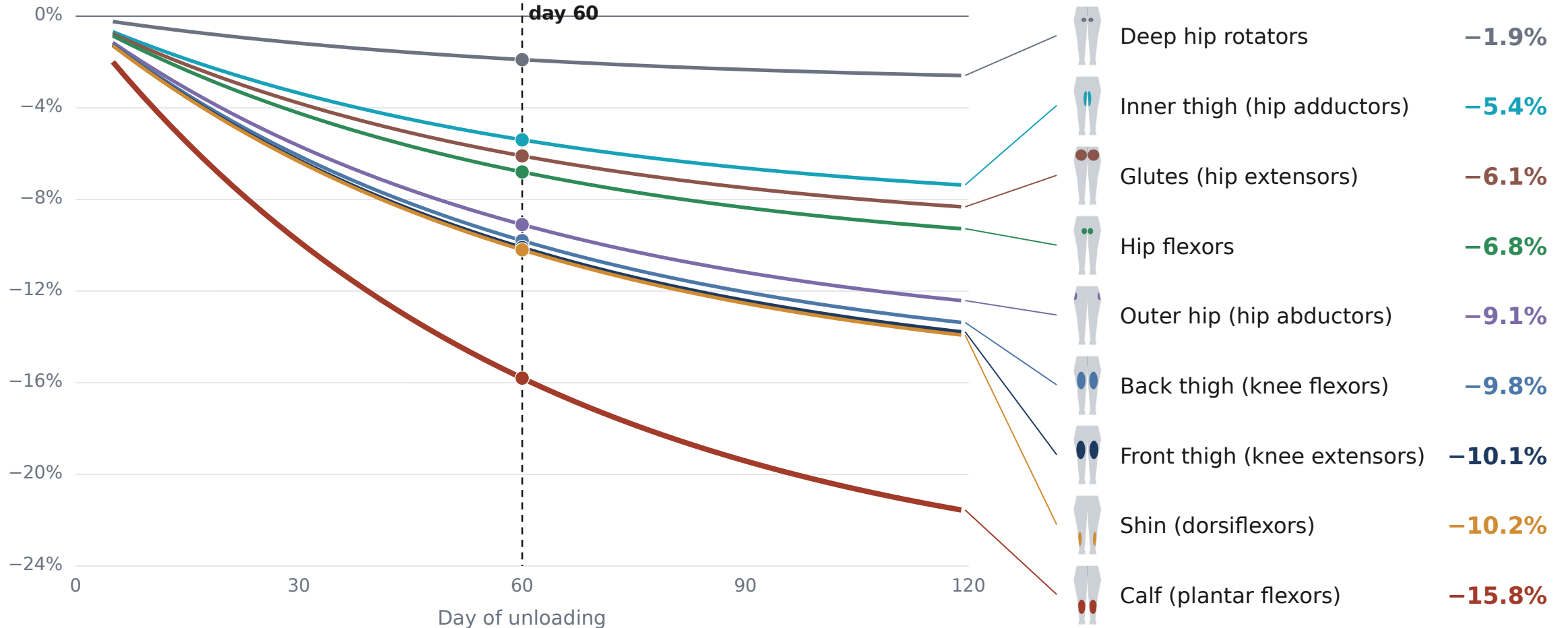
Imaging method and size measure

Method	n	Size measure	n
MRI	655	Volume	594
DXA	37	Cross-sectional area	100
CT	30	Lean mass	37
Ultrasound	20	Thickness	11
Total	742	Total	742

742 observations are not 742 independent participants: one campaign gives rows for several muscles, groups and scan days.

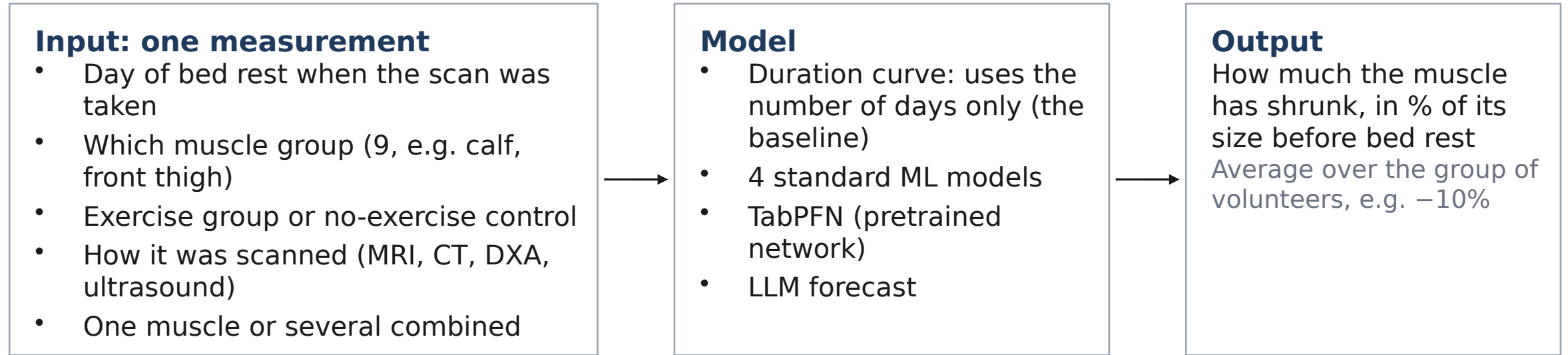
Are all muscles affected the same?

Change in muscle size



Lines: one shared time course (saturating, $\tau = 60$ days) scaled to each group's estimate at day 60. Control groups, days 5 to 119.

Every model is tested on a campaign it has not seen



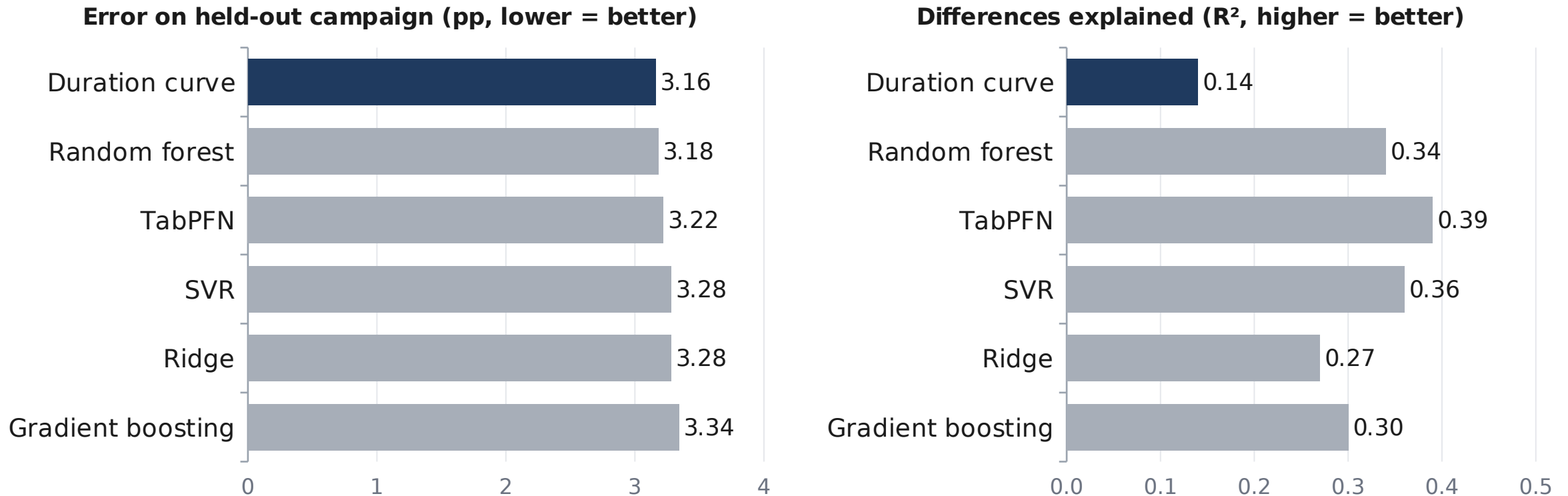
Testing: hold out one campaign, train on the rest, repeat 32 times



Grey: 31 training campaigns. Red: the held-out test campaign. Each campaign is held out once.

- Why campaigns, not measurements: the 346 rows used for modelling come from 32 of the 36 campaigns, and rows of one campaign share volunteers.
- Score: average error in percentage points (pp), each campaign counted once. Forecast -10%, measured -13%: error 3 pp.

No ML model has lower error than the duration curve



- R^2 is the share of the differences between measurements that a model explains: 0 is none, 1 is all. The ML models explain more (0.27 to 0.39 vs 0.14) because they learn which muscles shrink more.
- Their error on a new campaign is still no lower. Most of the error left is a shift that affects a whole campaign (its scanner, volunteers, protocol), and none of our inputs describe it.

LLM forecast: what is sent and what comes back

Request (one per measurement)

participants

men only, mean age 35, n = 11

protocol

head-down tilt bed rest, 21 days, control group

measurement

vastus lateralis, MRI cross-sectional area

target

day 21 of bed rest

typical_curve_other_campaigns

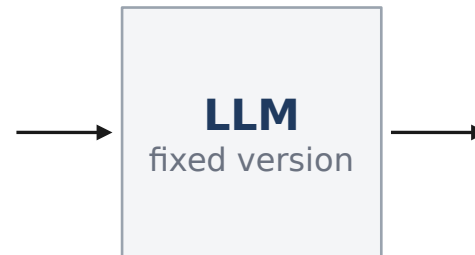
duration curve fitted on the other 31 campaigns

observations_other_campaigns

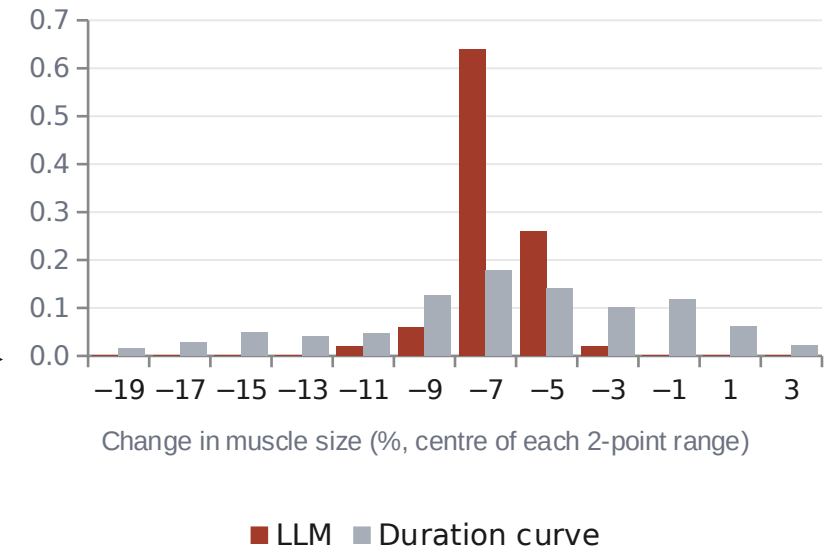
rows from the other 31 campaigns, most similar first, up to about 26,000 tokens

question

Which range will the change fall in? 19 options: below -30%, 2-point ranges up to +4%, +4% and above



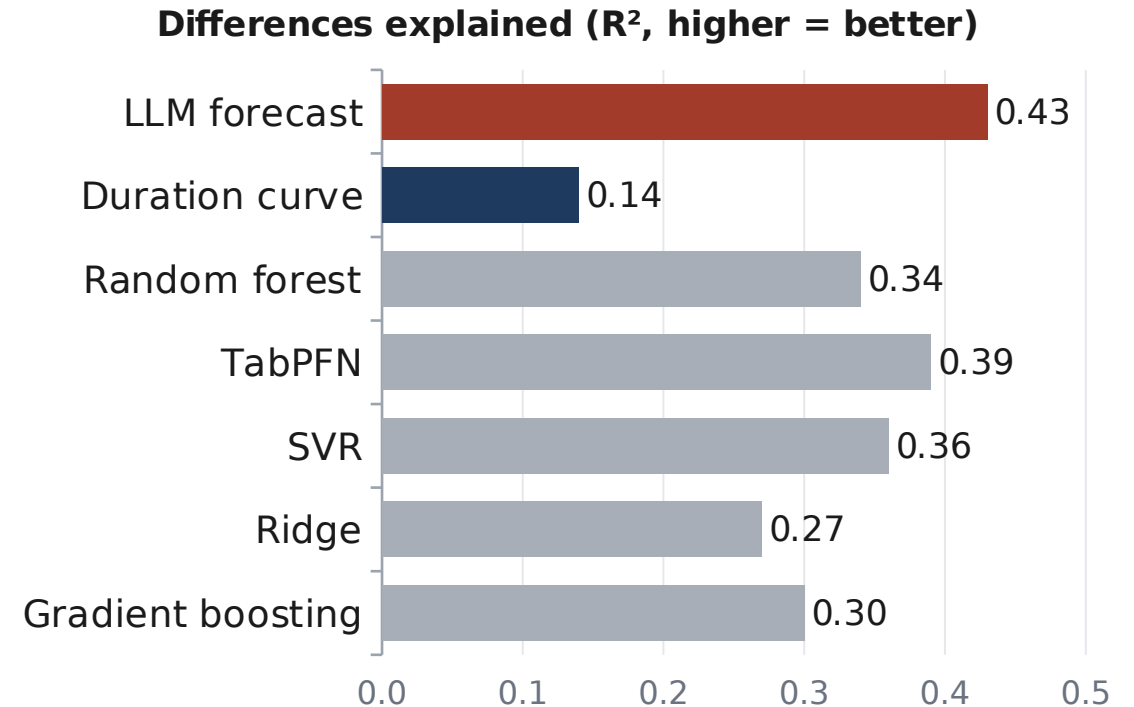
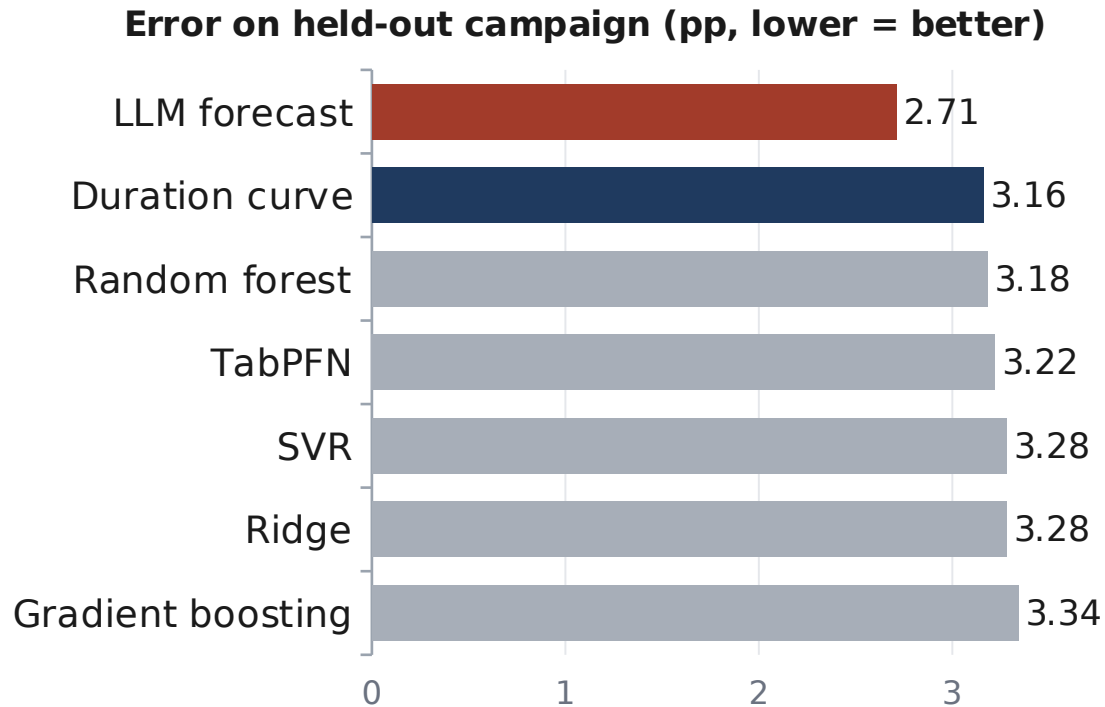
Answer: probability per range



- Our code turns the probabilities into one forecast, their weighted average: -6.6%
- Measured: -9.4%, an error of 2.8 pp. This is a typical row: half of the LLM's errors are smaller.

Never sent (checked on every request): the held-out campaign's measurements, and any paper, author or campaign name. All 430 answers are saved, so every result can be rebuilt without calling the model again.

The LLM forecast has the lowest error of all models



- Same measurements, scored the same way: error 2.71 vs 3.13 pp, so 0.42 pp smaller. Closer than the curve in 20 of 32 campaigns.
- 95% confidence interval of that difference: 0.13 to 0.73 pp. The whole range is above zero, so it is unlikely to be luck.

The curve shows 3.13 pp here, not 3.16, because the comparison scores it exactly as it scores the LLM.

Validity checks on the LLM forecast

Gain = how much smaller the LLM's error is than the curve's, in pp (full result: 0.42). Each check and its pass rule were fixed before running it.

Does it recognise a known campaign?

- Everything identifying a campaign removed: gain still 0.30 pp, interval above zero
- Asked to choose the campaign's name from a list of options: right 19% of the time, where guessing gives 8%

Does it use the other campaigns' data?

- Their values shuffled: worse than the curve (gain -1.25 pp)
- None of their data given: error 5.85 vs 3.13 pp

Is it sensitive to presentation?

- Other campaigns' rows in a different order: error changes 3%
- Answer ranges moved by 1 point: error changes 1%

Is it reproducible?

- 20 identical requests resent: forecasts differ by 0.17 pp on average, well below the 0.42 pp gain

Limitations of this result

- Added after the ML result was known; not in the original plan
- Overconfident ranges: its 80% ranges hold the measured value 62% of the time, so we report only its single-number forecast
- Given a campaign's own earlier scans, it does not use them

Conclusions, limitations and further work

Conclusions

- With 32 independent campaigns, no ML model forecasts better than a curve based on days alone (best 3.18 vs 3.16 pp error).
- An LLM that reads each campaign's description and the other campaigns' data cuts the error to 2.71 pp (curve 3.13 pp); the gain remains when everything identifying a campaign is hidden.

Limitations

- Only 32 independent campaigns
- No data beyond day 119
- Mostly young men (347 of 425 measurements)
- Group averages, not individual people
- Bed rest imitates spaceflight but is not the same

Further work

- Fix the LLM's overconfident ranges
- Check results without the largest campaign, and with MRI scans only
- Extend the forecast to a 180-day mission, with an uncertainty range
- Link the muscles: predict one muscle's loss from another's, and weight them into an estimate of strength loss

Backup: is the dataset good enough?

Did you miss studies?

- Four sources searched from 2013 on, plus 10 older or open-data studies (sources and queries on the next slide)
- Not covered: Embase (no access) and 1,023 records not screened in time

One campaign dominates the data

- MEDES 90-day: 40% of all rows, a quarter of the modelling data
- Scoring gives each campaign one vote; the re-run without it is still to do

Can MRI, CT, DXA and ultrasound be mixed?

- Each method gets its own term: DXA and ultrasound thickness show 5 to 6 pp less loss than CT; MRI volume is within 0.5 pp
- The MRI-only re-run is still to do

How reliable is the extraction?

- Every value is traced to its paper and page, table or figure
- 674 of 742 rows high confidence; 19 read off figures; no independent second extraction

Backup: how did we search the literature?

Sources (searched 4 September 2026)

Source	Records
PubMed	1,412
Scopus	1,757
Web of Science Core Collection	1,876
NASA Technical Reports Server	686
Earlier studies and NASA open data	10
Total	5,741

- Journal databases: publications from 2013 onwards
- Older work enters through studies we held before the search (9) and one NASA open-data campaign
- NASA reports: five short keyword queries, e.g. "bed rest" "muscle volume"
- Not searched: Embase (no institutional access), Cochrane CENTRAL, trial registries, citation chasing

Query: two required blocks

Unloading model

bed rest, head-down tilt, dry immersion, limb suspension, simulated microgravity, disuse

AND

Muscle outcome

atrophy, muscle volume, mass, size, cross-sectional area, lean mass, thickness, main leg muscles

NOT

Excluded

rat, mouse, rodent, hindlimb

Web of Science string, verbatim

```
TS=((("bed rest" OR bedrest OR "head-down tilt" OR "head down bed rest" OR HDBR  
OR antiorthostatic OR hypokinesia OR hypodynamia OR "dry immersion"  
OR "limb suspension" OR ULLS OR "simulated microgravity"  
OR "microgravity analog*" OR "spaceflight analog*" OR "mechanical unloading"  
OR "muscle unloading" OR disuse)  
AND  
(atroph* OR "muscle volume" OR "muscle mass" OR "muscle size"  
OR "cross-sectional area" OR PCSA OR "lean mass" OR "muscle thickness"  
OR "muscle wasting" OR deconditioning OR soleus OR gastrocnemius  
OR "triceps surae" OR quadriceps OR "vastus lateralis" OR "knee extensor"))  
NOT TS=(rat OR rats OR mice OR mouse OR rodent OR hindlimb OR "hind limb")
```

Backup: is the LLM gain real, and does it matter?

Is 0.42 pp worth anything?

- It is 13% lower error than the curve (3.13 to 2.71 pp)
- Small; we present it as a secondary finding

20 of 32 campaigns is close to a coin flip

- Counting wins only: $p = 0.22$ (sign test)*
- Using the size of each win: mean gain 0.42 pp, 95% CI 0.13 to 0.73

Does it rest on a few campaigns?

- 3 long MRI campaigns (NASA SPRINT, Berlin BBR2-2, MEDES LTBR) give 56% of the gain; without them 0.20 pp, CI -0.01 to 0.43^*
- Only SPRINT is recognised, and with campaign details removed the LLM still wins there (3.79 vs 5.92 pp)

Did it read these papers in training? Was it added after the fact?

- Training exposure cannot be ruled out by any test; hiding campaign details and the naming test are the strongest checks available
- Added after the ML result, but every check and target was committed before the first answer

** Our calculation from results/forecast_predictions.csv; not in the report.*

Backup: why did ML fail, and was the comparison fair?

Did the ML models get a fair chance?

- Tuned inside each training fold, grouped by campaign; TabPFN needs no tuning
- All models saw the same inputs, folds and scoring

Why not add age, sex or countermeasure type?

- 32 campaigns leave room for about 3 campaign-level inputs (about 10 campaigns per input)
- Age and sex are fixed within a campaign (347 of 425 rows men-only), so they act as a campaign label; 126 countermeasure rows spread over 9 types

Is the duration curve a straw man?

- Ridge regression uses days, muscle and every other input: 3.28 vs 3.16 pp
- Knowing the muscle raises R^2 but not the error on a new campaign

Would a nearest-neighbour method match the LLM?

- Not tested. If the gain comes from picking similar rows, this is the next control to run

Backup: can this be used to plan a Mars mission?

A Mars transit is about 180 days; the data stop at 119

- Only one campaign reaches 119 days
- A 180-day estimate is planned from the duration curve only, with an uncertainty range, labelled as an assumption

Which countermeasure works? What about individuals?

- Countermeasure groups lose 3.7 pp less (95% CI 1.5 to 5.9), all types pooled
- Every row is a group average; nothing about individual risk

Bed rest is not spaceflight

- Only 14 spaceflight rows, 8 of them one back muscle: too few to test transfer

Can anyone reproduce an LLM result?

- Fixed model version; all 430 answers saved, so every number rebuilds without the model
- Identical requests differ by 0.17 pp on average